black_scholes_model

April 25, 2026

1 Black–Scholes Modeling Layer

This notebook implements the modeling lead deliverable for the delta-hedging project. It loads the cleaned project data, computes Black–Scholes theoretical option values and deltas over each option episode, compares implied-volatility and realized-volatility inputs, validates the results, and saves a clean panel for the hedging-simulation teammate.

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path
from scipy.stats import norm

pd.set_option("display.max_columns", None)
pd.set_option("display.width", 120)
```

1.1 1. Portable project paths

[ ]:
```
PROJECT_ROOT = Path.cwd().resolve()

while PROJECT_ROOT != PROJECT_ROOT.parent and not (PROJECT_ROOT / "data" /␣
↪"output").exists():
    PROJECT_ROOT = PROJECT_ROOT.parent

DATA_PATH = PROJECT_ROOT / "data" / "output"

if not DATA_PATH.exists():
    raise FileNotFoundError(
        f"Could not find data/output from current directory. Current resolved␣
↪root: {PROJECT_ROOT}"
    )

print("Project root:", PROJECT_ROOT)
print("Data path:", DATA_PATH)
print("Available files:")
for f in sorted(DATA_PATH.iterdir()):
    print(" -", f.name)
```

```
Project root: C:\Users\elain\Downloads\5010-Project
Data path: C:\Users\elain\Downloads\5010-Project\data\output
Available files:
 - data_quality_report.md
 - dividends.parquet
 - episodes.parquet
 - option_chain_snapshot.parquet
 - risk_free.parquet
 - stock_intraday_15m.parquet
 - stock_intraday_1h.parquet
 - stock_prices.parquet
 - vix.parquet
```

1.2 2. Load cleaned data

Main inputs: - `episodes.parquet`: the 12 option experiments. - `stock_prices.parquet`: SPY daily prices and realized-volatility columns. - `risk_free.parquet`: daily risk-free rate in decimal form.

[3]:
```
episodes = pd.read_parquet(DATA_PATH / "episodes.parquet")
prices = pd.read_parquet(DATA_PATH / "stock_prices.parquet")
rf = pd.read_parquet(DATA_PATH / "risk_free.parquet")

print("Episodes:", episodes.shape)
print("Prices:", prices.shape)
print("Risk-free:", rf.shape)

display(episodes.head())
display(prices.head())
display(rf.head())
```

```
Episodes: (12, 18)
Prices: (1583, 11)
Risk-free: (1583, 2)

         episode_id  regime ticker cp_flag moneyness inception_date          S0 ␣
 ↪strike expiration  T_days   T_years  \
0    SPY_calm_C_ATM    calm    SPY       C       ATM     2021-03-01  389.579987 ␣
 ↪ 390.0 2021-05-28      63  0.250000
1    SPY_calm_C_OTM    calm    SPY       C       OTM     2021-03-01  389.579987 ␣
 ↪ 409.0 2021-05-28      63  0.250000
2    SPY_calm_P_ATM    calm    SPY       P       ATM     2021-03-01  389.579987 ␣
 ↪ 390.0 2021-05-28      63  0.250000
3  SPY_stress_C_ATM  stress    SPY       C       ATM     2020-02-03  324.119995 ␣
 ↪ 324.0 2020-05-01      62  0.246032
4  SPY_stress_C_OTM  stress    SPY       C       OTM     2020-02-03  324.119995 ␣
 ↪ 340.0 2020-05-01      62  0.246032
```

```
   iv_inception_vix  iv_inception_vix3m  r_inception  div_amount_in_window␣
 ↪div_ex_dates  pv_div_inception  q_inception
0            0.2335              0.2678      0.00028                 1.278  ␣
 ↪2021-03-19          1.277982     0.013143
1            0.2335              0.2678      0.00028                 1.278  ␣
 ↪2021-03-19          1.277982     0.013143
2            0.2335              0.2678      0.00028                 1.278  ␣
 ↪2021-03-19          1.277982     0.013143
3            0.1797              0.1804      0.01518                 1.406  ␣
 ↪2020-03-20          1.403313     0.017636
4            0.1797              0.1804      0.01518                 1.406  ␣
 ↪2020-03-20          1.403313     0.017636

        date ticker        open        high         low       close    volume  ␣
 ↪adj_close    logret  rv_20d  rv_60d
0 2020-01-02    SPY  323.540009  324.890015  322.529999  324.869995  59151200 ␣
 ↪296.888123       NaN     NaN     NaN
1 2020-01-03    SPY  321.160004  323.640015  321.100006  322.410004  77709700 ␣
 ↪294.640045 -0.007601     NaN     NaN
2 2020-01-06    SPY  320.489990  323.730011  320.359985  323.640015  55653900 ␣
 ↪295.764160  0.003808     NaN     NaN
3 2020-01-07    SPY  323.019989  323.540009  322.239990  322.730011  40496400 ␣
 ↪294.932526 -0.002816     NaN     NaN
4 2020-01-08    SPY  322.940002  325.779999  322.670013  324.450012  68296000 ␣
 ↪296.504425  0.005315     NaN     NaN

        date        r
0 2020-01-02  0.01495
1 2020-01-03  0.01473
2 2020-01-06  0.01488
3 2020-01-07  0.01500
4 2020-01-08  0.01493
```

[4]:
```
print("Episode columns:")
print(episodes.columns.tolist())

print("Price columns:")
print(prices.columns.tolist())

print("Risk-free columns:")
print(rf.columns.tolist())
```

```
Episode columns:
['episode_id', 'regime', 'ticker', 'cp_flag', 'moneyness', 'inception_date',
'S0', 'strike', 'expiration', 'T_days', 'T_years', 'iv_inception_vix',
'iv_inception_vix3m', 'r_inception', 'div_amount_in_window', 'div_ex_dates',
'pv_div_inception', 'q_inception']
Price columns:
```

```
['date', 'ticker', 'open', 'high', 'low', 'close', 'volume', 'adj_close',
'logret', 'rv_20d', 'rv_60d']
Risk-free columns:
['date', 'r']
```

1.3 3. Prepare daily modeling data

Use the traded SPY `close` price for hedge marking, not the adjusted close. We merge risk-free rates onto trading days only, so SPY prices are not forward-filled into non-trading days.

[5]:
```
# Ensure datetime format
prices = prices.copy()
rf = rf.copy()
prices["date"] = pd.to_datetime(prices["date"])
rf["date"] = pd.to_datetime(rf["date"])

data = prices.merge(rf, on="date", how="left")

# Risk-free rates can be filled across available trading rows; do not fill SPY␣
↪prices into non-trading days.
data["r"] = data["r"].ffill().bfill()

PRICE_COL = "close"
RF_COL = "r"

required_data_cols = ["date", PRICE_COL, RF_COL, "rv_20d", "rv_60d"]
missing_data_cols = [c for c in required_data_cols if c not in data.columns]
if missing_data_cols:
    raise KeyError(f"Missing required columns from merged data:␣
↪{missing_data_cols}")

display(data.head())
```

```
        date ticker        open        high         low       close    volume  ␣
↪adj_close    logret  rv_20d  rv_60d  \
0 2020-01-02    SPY  323.540009  324.890015  322.529999  324.869995  59151200 ␣
↪296.888123       NaN     NaN     NaN
1 2020-01-03    SPY  321.160004  323.640015  321.100006  322.410004  77709700 ␣
↪294.640045 -0.007601     NaN     NaN
2 2020-01-06    SPY  320.489990  323.730011  320.359985  323.640015  55653900 ␣
↪295.764160  0.003808     NaN     NaN
3 2020-01-07    SPY  323.019989  323.540009  322.239990  322.730011  40496400 ␣
↪294.932526 -0.002816     NaN     NaN
4 2020-01-08    SPY  322.940002  325.779999  322.670013  324.450012  68296000 ␣
↪296.504425  0.005315     NaN     NaN

         r
0  0.01495
```

```
1  0.01473
2  0.01488
3  0.01500
4  0.01493
```

1.4 4. Black–Scholes functions

The formulas include a continuous dividend-yield proxy `q`. The functions are vector-safe and include basic numerical protection near maturity or near-zero volatility.

[6]:
```
def bs_d1(S, K, r, q, sigma, tau):
    """Black-Scholes d1 with numerical safeguards."""
    S = np.asarray(S, dtype=float)
    K = np.asarray(K, dtype=float)
    r = np.asarray(r, dtype=float)
    q = np.asarray(q, dtype=float)
    sigma = np.asarray(sigma, dtype=float)
    tau = np.asarray(tau, dtype=float)

    sigma = np.maximum(sigma, 1e-8)
    tau = np.maximum(tau, 1e-8)

    return (np.log(S / K) + (r - q + 0.5 * sigma**2) * tau) / (sigma * np.
 ↪sqrt(tau))


def bs_d2(d1, sigma, tau):
    sigma = np.asarray(sigma, dtype=float)
    tau = np.asarray(tau, dtype=float)
    sigma = np.maximum(sigma, 1e-8)
    tau = np.maximum(tau, 1e-8)
    return d1 - sigma * np.sqrt(tau)


def bs_price(S, K, r, q, sigma, tau, option_type="call"):
    """Black-Scholes price for European call or put with continuous dividend␣
 ↪yield."""
    d1 = bs_d1(S, K, r, q, sigma, tau)
    d2 = bs_d2(d1, sigma, tau)

    if option_type == "call":
        return S * np.exp(-q * tau) * norm.cdf(d1) - K * np.exp(-r * tau) *␣
 ↪norm.cdf(d2)
    elif option_type == "put":
        return K * np.exp(-r * tau) * norm.cdf(-d2) - S * np.exp(-q * tau) *␣
 ↪norm.cdf(-d1)
    else:
```

```
        raise ValueError("option_type must be 'call' or 'put'")


def bs_delta(S, K, r, q, sigma, tau, option_type="call"):
    """Black-Scholes delta for European call or put with continuous dividend␣
 ↪yield."""
    d1 = bs_d1(S, K, r, q, sigma, tau)

    if option_type == "call":
        delta = np.exp(-q * tau) * norm.cdf(d1)
    elif option_type == "put":
        delta = np.exp(-q * tau) * (norm.cdf(d1) - 1)
    else:
        raise ValueError("option_type must be 'call' or 'put'")

    return np.clip(delta, -1, 1)
```

1.5 5. Build Black–Scholes delta panel

This creates one row per option episode, per trading day, per volatility specification. Including both implied and realized volatility lets the project directly compare how volatility input choices change theoretical deltas and downstream hedging performance.

[7]:
```
# Core columns in episodes.parquet
EPISODE_ID_COL = "episode_id"
REGIME_COL = "regime"
START_COL = "inception_date"
END_COL = "expiration"
STRIKE_COL = "strike"
CP_COL = "cp_flag"
MONEYNESS_COL = "moneyness"
S0_COL = "S0"
R0_COL = "r_inception"
T_COL = "T_years"
DIV_COL = "div_amount_in_window"

required_episode_cols = [
    EPISODE_ID_COL, REGIME_COL, START_COL, END_COL, STRIKE_COL,
    CP_COL, MONEYNESS_COL, S0_COL, R0_COL, T_COL
]
missing_episode_cols = [c for c in required_episode_cols if c not in episodes.
 ↪columns]
if missing_episode_cols:
    raise KeyError(f"Missing required columns from episodes:␣
 ↪{missing_episode_cols}")
```

```
#compare short-tenor implied vol, 3-month implied vol, and historical realized␣
↪vols.
VOL_METHODS = ["iv_inception_vix", "iv_inception_vix3m", "rv_20d", "rv_60d"]

missing_vol_cols = [c for c in ["iv_inception_vix", "iv_inception_vix3m"] if c␣
↪not in episodes.columns]
if missing_vol_cols:
    raise KeyError(f"Missing implied-vol columns from episodes:␣
↪{missing_vol_cols}")

results = []

for vol_method in VOL_METHODS:
    for _, row in episodes.iterrows():
        episode_id = row[EPISODE_ID_COL]
        regime = row[REGIME_COL]
        start = pd.to_datetime(row[START_COL])
        end = pd.to_datetime(row[END_COL])
        K = float(row[STRIKE_COL])
        S0 = float(row[S0_COL])
        option_type = "call" if row[CP_COL] == "C" else "put"

        # Continuous-dividend-yield proxy based on expected dividend cash␣
↪amount during the option window.
        # If the column is not available or invalid, fall back to q = 0.
        if DIV_COL in episodes.columns and pd.notna(row[DIV_COL]) and␣
↪row[T_COL] > 0 and S0 > 0:
            q = float(row[DIV_COL]) / S0 / float(row[T_COL])
            q_source = "div_amount_in_window / S0 / T_years"
        else:
            q = 0.0
            q_source = "fallback_zero"

        df = data[(data["date"] >= start) & (data["date"] <= end)].copy()
        if df.empty:
            print(f"Warning: no trading-day data for {episode_id} from {start.
↪date()} to {end.date()}")
            continue

        for _, obs in df.iterrows():
            tau = (end - obs["date"]).days / 365.0
            if tau <= 0:
                continue

            S = float(obs[PRICE_COL])
```

```
            r_rate = float(obs[RF_COL]) if pd.notna(obs[RF_COL]) else␣
↪float(row[RO_COL])

            if vol_method == "iv_inception_vix":
                sigma = row["iv_inception_vix"]
            elif vol_method == "iv_inception_vix3m":
                sigma = row["iv_inception_vix3m"]
            elif vol_method == "rv_20d":
                sigma = obs["rv_20d"]
            elif vol_method == "rv_60d":
                sigma = obs["rv_60d"]
            else:
                raise ValueError(f"Unknown vol_method: {vol_method}")

            if pd.isna(sigma) or sigma <= 0:
                continue

            sigma = float(sigma)
            price = float(bs_price(S, K, r_rate, q, sigma, tau, option_type))
            delta = float(bs_delta(S, K, r_rate, q, sigma, tau, option_type))

            # Keep delta in valid theoretical range and remove tiny␣
↪floating-point noise near zero.
            delta = float(np.clip(delta, -1.0, 1.0))
            if abs(delta) < 1e-10:
                delta = 0.0

            results.append({
                "episode_id": episode_id,
                "regime": regime,
                "date": obs["date"],
                "S": S,
                "K": K,
                "tau": tau,
                "r": r_rate,
                "q": q,
                "sigma": sigma,
                "bs_price": price,
                "delta": delta,
                "option_type": option_type,
                "cp_flag": row[CP_COL],
                "moneyness": row[MONEYNESS_COL],
                "vol_method": vol_method,
                "q_source": q_source,
                "inception_date": start,
                "expiration": end,
                "S0": S0
```

```
            })

bs_panel = pd.DataFrame(results).sort_values(["vol_method", "episode_id",␣
 ↪"date"]).reset_index(drop=True)

print("Black-Scholes panel shape:", bs_panel.shape)
display(bs_panel.head())
```

```
Black-Scholes panel shape: (2847, 19)

       episode_id regime       date           S      K       tau        r       ␣
 ↪ q   sigma   bs_price     delta  \
0  SPY_calm_C_ATM   calm 2021-03-01  389.579987  390.0  0.241096  0.00028  0.
 ↪013122  0.2335  16.989992  0.506741
1  SPY_calm_C_ATM   calm 2021-03-02  386.540009  390.0  0.238356  0.00035  0.
 ↪013122  0.2335  15.399808  0.479406
2  SPY_calm_C_ATM   calm 2021-03-03  381.420013  390.0  0.235616  0.00035  0.
 ↪013122  0.2335  12.971057  0.432680
3  SPY_calm_C_ATM   calm 2021-03-04  376.700012  390.0  0.232877  0.00028  0.
 ↪013122  0.2335  10.937231  0.389266
4  SPY_calm_C_ATM   calm 2021-03-05  383.630005  390.0  0.230137  0.00028  0.
 ↪013122  0.2335  13.758139  0.451908

  option_type cp_flag moneyness        vol_method                            ␣
 ↪q_source inception_date expiration  \
0        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
1        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
2        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
3        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
4        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28

           S0
0  389.579987
1  389.579987
2  389.579987
3  389.579987
4  389.579987
```

1.6 6. Quality checks

These checks confirm the panel is ready for handoff: no missing model outputs, correct delta ranges, and sensible episode coverage.

[8]:
```
print("Shape:", bs_panel.shape)
print("Rows by volatility method:")
display(bs_panel.groupby("vol_method").size().rename("n_rows").to_frame())

print("Missing values:")
display(bs_panel.isna().sum().to_frame("missing_count"))

print("Delta range by option type and volatility method:")
display(
    bs_panel.groupby(["vol_method", "option_type"])["delta"]
    .describe()
)
```

```
Shape: (2847, 19)
Rows by volatility method:

                    n_rows
vol_method
iv_inception_vix       741
iv_inception_vix3m     741
rv_20d                 741
rv_60d                 624

Missing values:

                missing_count
episode_id                  0
regime                      0
date                        0
S                           0
K                           0
tau                         0
r                           0
q                           0
sigma                       0
bs_price                    0
delta                       0
option_type                 0
cp_flag                     0
moneyness                   0
vol_method                  0
q_source                    0
inception_date              0
expiration                  0
S0                          0

Delta range by option type and volatility method:

                                count      mean       std       min       25%
 ↪   50%       75%           max
```

```
vol_method         option_type
iv_inception_vix   call         494.0  0.329537  0.294446  0.000000  0.009465  0.
 ↪302847  0.567085  9.999640e-01
                   put          247.0 -0.576541  0.325879 -0.999961 -0.963870 -0.
 ↪494723 -0.310319 -1.530698e-09
iv_inception_vix3m call         494.0  0.336787  0.283344  0.000000  0.029327  0.
 ↪326483  0.562621  9.999639e-01
                   put          247.0 -0.576027  0.312627 -0.999961 -0.932776 -0.
 ↪492196 -0.327421 -1.166364e-07
rv_20d             call         494.0  0.368143  0.299337  0.000000  0.127474  0.
 ↪272003  0.585284  9.999641e-01
                   put          247.0 -0.513411  0.315327 -0.999961 -0.798464 -0.
 ↪502336 -0.267798  0.000000e+00
rv_60d             call         416.0  0.359436  0.314924  0.000000  0.077412  0.
 ↪253607  0.596761  9.999641e-01
                   put          208.0 -0.512943  0.331967 -0.999961 -0.873176 -0.
 ↪496809 -0.227109  0.000000e+00
```

[9]:
```python
# Calls should be between 0 and 1; puts should be between -1 and 0.
bad_calls = bs_panel[
    (bs_panel["option_type"] == "call") &
    ((bs_panel["delta"] < -1e-12) | (bs_panel["delta"] > 1 + 1e-12))
]

bad_puts = bs_panel[
    (bs_panel["option_type"] == "put") &
    ((bs_panel["delta"] < -1 - 1e-12) | (bs_panel["delta"] > 1e-12))
]

print("Bad call rows:", len(bad_calls))
print("Bad put rows:", len(bad_puts))

if len(bad_calls) > 0:
    display(bad_calls.head())
if len(bad_puts) > 0:
    display(bad_puts.head())
```

```
Bad call rows: 0
Bad put rows: 0
```

[10]:
```python
episode_summary = bs_panel.groupby(["vol_method", "episode_id"]).agg(
    n_days=("date", "count"),
    first_date=("date", "min"),
    last_date=("date", "max"),
    min_tau=("tau", "min"),
    max_tau=("tau", "max"),
    min_delta=("delta", "min"),
```

```
    max_delta=("delta", "max"),
    first_price=("bs_price", "first"),
    last_price=("bs_price", "last"),
    mean_sigma=("sigma", "mean")
)

display(episode_summary)
```

```
                                    n_days first_date  last_date   min_tau ␣
↪max_tau     min_delta     max_delta  \
vol_method          episode_id
iv_inception_vix    SPY_calm_C_ATM        63 2021-03-01 2021-05-27  0.002740  0.
↪241096  3.892658e-01  9.999640e-01
                    SPY_calm_C_OTM        63 2021-03-01 2021-05-27  0.002740  0.
↪241096  2.411383e-01  9.791005e-01
                    SPY_calm_P_ATM        63 2021-03-01 2021-05-27  0.002740  0.
↪241096 -6.076831e-01 -1.530698e-09
                    SPY_highvol_C_ATM     62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  5.430961e-01
                    SPY_highvol_C_OTM     62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  3.430716e-01
                    SPY_highvol_P_ATM     62 2022-04-01 2022-06-30  0.002740  0.
↪249315 -9.999612e-01 -4.534977e-01
                    SPY_recent_C_ATM      60 2025-10-01 2025-12-24  0.005479  0.
↪235616  3.501367e-01  9.969460e-01
                    SPY_recent_C_OTM      60 2025-10-01 2025-12-24  0.005479  0.
↪235616  4.385506e-02  4.079310e-01
                    SPY_recent_P_ATM      60 2025-10-01 2025-12-24  0.005479  0.
↪235616 -6.486290e-01 -2.985381e-03
                    SPY_stress_C_ATM      62 2020-02-03 2020-04-30  0.002740  0.
↪241096  0.000000e+00  7.155668e-01
                    SPY_stress_C_OTM      62 2020-02-03 2020-04-30  0.002740  0.
↪241096  0.000000e+00  4.875876e-01
                    SPY_stress_P_ATM      62 2020-02-03 2020-04-30  0.002740  0.
↪241096 -9.999517e-01 -2.809612e-01
iv_inception_vix3m SPY_calm_C_ATM         63 2021-03-01 2021-05-27  0.002740  0.
↪241096  4.089553e-01  9.999639e-01
                    SPY_calm_C_OTM        63 2021-03-01 2021-05-27  0.002740  0.
↪241096  2.750481e-01  9.621893e-01
                    SPY_calm_P_ATM        63 2021-03-01 2021-05-27  0.002740  0.
↪241096 -5.879936e-01 -1.166364e-07
                    SPY_highvol_C_ATM     62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  5.427009e-01
                    SPY_highvol_C_OTM     62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  3.790008e-01
```

```
                    SPY_highvol_P_ATM      62 2022-04-01 2022-06-30  0.002740  0.
↪249315 -9.999612e-01 -4.538929e-01
                    SPY_recent_C_ATM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616  3.717501e-01  9.914866e-01
                    SPY_recent_C_OTM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616  6.954542e-02  4.233430e-01
                    SPY_recent_P_ATM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616 -6.270156e-01 -8.444754e-03
                    SPY_stress_C_ATM       62 2020-02-03 2020-04-30  0.002740  0.
↪241096  0.000000e+00  7.149172e-01
                    SPY_stress_C_OTM       62 2020-02-03 2020-04-30  0.002740  0.
↪241096  0.000000e+00  4.877524e-01
                    SPY_stress_P_ATM       62 2020-02-03 2020-04-30  0.002740  0.
↪241096 -9.999517e-01 -2.816108e-01
rv_20d              SPY_calm_C_ATM         63 2021-03-01 2021-05-27  0.002740  0.
↪241096  3.291647e-01  9.999641e-01
                    SPY_calm_C_OTM         63 2021-03-01 2021-05-27  0.002740  0.
↪241096  1.478472e-01  9.995269e-01
                    SPY_calm_P_ATM         63 2021-03-01 2021-05-27  0.002740  0.
↪241096 -6.677842e-01  0.000000e+00
                    SPY_highvol_C_ATM      62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  5.430871e-01
                    SPY_highvol_C_OTM      62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  3.451556e-01
                    SPY_highvol_P_ATM      62 2022-04-01 2022-06-30  0.002740  0.
↪249315 -9.999612e-01 -4.535067e-01
                    SPY_recent_C_ATM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616  3.264192e-01  9.999313e-01
                    SPY_recent_C_OTM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616  4.919664e-03  3.903330e-01
                    SPY_recent_P_ATM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616 -6.723465e-01 -1.204497e-07
                    SPY_stress_C_ATM       62 2020-02-03 2020-04-30  0.002740  0.
↪241096  4.855939e-09  7.699766e-01
                    SPY_stress_C_OTM       62 2020-02-03 2020-04-30  0.002740  0.
↪241096  0.000000e+00  4.746858e-01
                    SPY_stress_P_ATM       62 2020-02-03 2020-04-30  0.002740  0.
↪241096 -9.999517e-01 -2.265515e-01
rv_60d              SPY_calm_C_ATM         63 2021-03-01 2021-05-27  0.002740  0.
↪241096  3.084654e-01  9.999641e-01
                    SPY_calm_C_OTM         63 2021-03-01 2021-05-27  0.002740  0.
↪241096  1.206087e-01  9.998357e-01
                    SPY_calm_P_ATM         63 2021-03-01 2021-05-27  0.002740  0.
↪241096 -6.884835e-01  0.000000e+00
                    SPY_highvol_C_ATM      62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  5.426830e-01
```

```
                  SPY_highvol_C_OTM      62 2022-04-01 2022-06-30  0.002740  0.
↪249315  0.000000e+00  3.594163e-01
                  SPY_highvol_P_ATM      62 2022-04-01 2022-06-30  0.002740  0.
↪249315 -9.999612e-01 -4.539108e-01
                  SPY_recent_C_ATM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616  2.956427e-01  9.996865e-01
                  SPY_recent_C_OTM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616  1.424341e-02  3.454599e-01
                  SPY_recent_P_ATM       60 2025-10-01 2025-12-24  0.005479  0.
↪235616 -7.031229e-01 -2.448863e-04
                  SPY_stress_C_ATM       23 2020-03-30 2020-04-30  0.002740  0.
↪087671  3.336636e-04  1.670404e-01
                  SPY_stress_C_OTM       23 2020-03-30 2020-04-30  0.002740  0.
↪087671  4.532381e-07  9.605920e-02
                  SPY_stress_P_ATM       23 2020-03-30 2020-04-30  0.002740  0.
↪087671 -9.996180e-01 -8.321387e-01

                                     first_price     last_price  mean_sigma
vol_method         episode_id
iv_inception_vix   SPY_calm_C_ATM       16.989992   2.927499e+01    0.233500
                   SPY_calm_C_OTM        9.735632   1.031462e+01    0.233500
                   SPY_calm_P_ATM       18.614213   1.259206e-09    0.233500
                   SPY_highvol_C_ATM    17.112695   7.195427e-72    0.196300
                   SPY_highvol_C_OTM     8.556925  2.171826e-114    0.196300
                   SPY_highvol_P_ATM    18.223564   7.574477e+01    0.196300
                   SPY_recent_C_ATM     23.277283   2.247027e+01    0.162900
                   SPY_recent_C_OTM      9.963393   3.260133e-01    0.162900
                   SPY_recent_P_ATM     18.772231   7.526795e-03    0.162900
                   SPY_stress_C_ATM     11.322877   4.208629e-32    0.179700
                   SPY_stress_C_OTM      5.340996   5.857322e-64    0.179700
                   SPY_stress_P_ATM     11.394132   3.353322e+01    0.179700
iv_inception_vix3m SPY_calm_C_ATM       19.598268   2.927499e+01    0.267800
                   SPY_calm_C_OTM       12.181216   1.036430e+01    0.267800
                   SPY_calm_P_ATM       21.222490   1.244372e-07    0.267800
                   SPY_highvol_C_ATM    21.228070   5.244276e-48    0.242100
                   SPY_highvol_C_OTM    12.322135   5.002977e-76    0.242100
                   SPY_highvol_P_ATM    22.338939   7.574477e+01    0.242100
                   SPY_recent_C_ATM     26.441336   2.248981e+01    0.187600
                   SPY_recent_C_OTM     12.827760   5.502741e-01    0.187600
                   SPY_recent_P_ATM     21.936284   2.706620e-02    0.187600
                   SPY_stress_C_ATM     11.367095   7.178977e-32    0.180400
                   SPY_stress_C_OTM      5.379983   1.754146e-63    0.180400
                   SPY_stress_P_ATM     11.438349   3.353322e+01    0.180400
rv_20d             SPY_calm_C_ATM       11.490678   2.927499e+01    0.139658
                   SPY_calm_C_OTM        4.869216   1.027535e+01    0.139658
                   SPY_calm_P_ATM       13.114899   4.692679e-23    0.139658
                   SPY_highvol_C_ATM    20.145499   5.405250e-31    0.270310
```

```
                    SPY_highvol_C_OTM    11.315094   9.713121e-49    0.270310
                    SPY_highvol_P_ATM    21.256368   7.574477e+01    0.270310
                    SPY_recent_C_ATM     10.602786   2.246275e+01    0.124080
                    SPY_recent_C_OTM      0.779960   6.858049e-03    0.124080
                    SPY_recent_P_ATM      6.097734   9.671521e-08    0.124080
                    SPY_stress_C_ATM      7.574237   4.414470e-09    0.506973
                    SPY_stress_C_OTM      2.269867   4.327233e-17    0.506973
                    SPY_stress_P_ATM      7.645491   3.353322e+01    0.506973
rv_60d              SPY_calm_C_ATM        9.952502   2.927499e+01    0.148683
                    SPY_calm_C_OTM        3.633256   1.027508e+01    0.148683
                    SPY_calm_P_ATM       11.576723   2.120096e-27    0.148683
                    SPY_highvol_C_ATM    18.742604   5.511565e-34    0.251279
                    SPY_highvol_C_OTM    10.026315   1.568494e-53    0.251279
                    SPY_highvol_P_ATM    19.853473   7.574477e+01    0.251279
                    SPY_recent_C_ATM     13.557498   2.246315e+01    0.113069
                    SPY_recent_C_OTM      2.310081   1.087350e-01    0.113069
                    SPY_recent_P_ATM      9.052446   4.060167e-04    0.113069
                    SPY_stress_C_ATM      2.258222   7.900477e-04    0.595565
                    SPY_stress_C_OTM      1.215666   7.925143e-07    0.595565
                    SPY_stress_P_ATM     65.008674   3.353401e+01    0.595565
```

1.7 7. Report resources

[11]:
```python
import matplotlib.pyplot as plt

FIG_DIR = DATA_PATH / "report_figures"
TABLE_DIR = DATA_PATH / "report_tables"
FIG_DIR.mkdir(exist_ok=True)
TABLE_DIR.mkdir(exist_ok=True)

vol_label_map = {
    "iv_inception_vix": "VIX-Implied Volatility",
    "iv_inception_vix3m": "VIX3M-Implied Volatility",
    "rv_20d": "20-Day Realized Volatility",
    "rv_60d": "60-Day Realized Volatility"
}

sample_id = "SPY_calm_C_ATM"
sample = bs_panel[bs_panel["episode_id"] == sample_id].copy()
sample_info = sample.iloc[0]

print("Figure directory:", FIG_DIR)
print("Table directory:", TABLE_DIR)
print("Sample episode used for report figures:", sample_id)
```

```
Figure directory:
C:\Users\elain\Downloads\5010-Project\data\output\report_figures
Table directory: C:\Users\elain\Downloads\5010-Project\data\output\report_tables
```

Sample episode used for report figures: SPY_calm_C_ATM

[12]:
```
vol_table = pd.DataFrame({
    "Volatility Input": [
        "VIX-Implied Volatility",
        "VIX3M-Implied Volatility",
        "20-Day Realized Volatility",
        "60-Day Realized Volatility"
    ],
    "Variable": [
        "iv_inception_vix",
        "iv_inception_vix3m",
        "rv_20d",
        "rv_60d"
    ],
    "Construction": [
        "30-day implied volatility fixed at inception",
        "3-month implied volatility fixed at inception",
        "Rolling realized volatility from daily returns",
        "Rolling realized volatility from daily returns"
    ],
    "Time Variation": [
        "Fixed within episode",
        "Fixed within episode",
        "Updated daily",
        "Updated daily"
    ]
})

display(vol_table)

vol_table.to_csv(TABLE_DIR / "table_volatility_inputs.csv", index=False)

fig, ax = plt.subplots(figsize=(14, 2.8))
ax.axis("off")
tbl = ax.table(
    cellText=vol_table.values,
    colLabels=vol_table.columns,
    loc="center",
    cellLoc="center"
)
tbl.auto_set_font_size(False)
tbl.set_fontsize(10)
tbl.scale(1, 1.6)
plt.tight_layout()
plt.savefig(FIG_DIR / "table_volatility_inputs.png", dpi=300,␣
 ↪bbox_inches="tight")
```

```
plt.show()
plt.close()
```

```
            Volatility Input            Variable                                 ␣
↪    Construction  \
0      VIX-Implied Volatility    iv_inception_vix    30-day implied volatility␣
↪fixed at inception
1    VIX3M-Implied Volatility  iv_inception_vix3m   3-month implied volatility␣
↪fixed at inception
2  20-Day Realized Volatility              rv_20d  Rolling realized volatility␣
↪from daily returns
3  60-Day Realized Volatility              rv_60d  Rolling realized volatility␣
↪from daily returns

        Time Variation
0  Fixed within episode
1  Fixed within episode
2         Updated daily
3         Updated daily
```

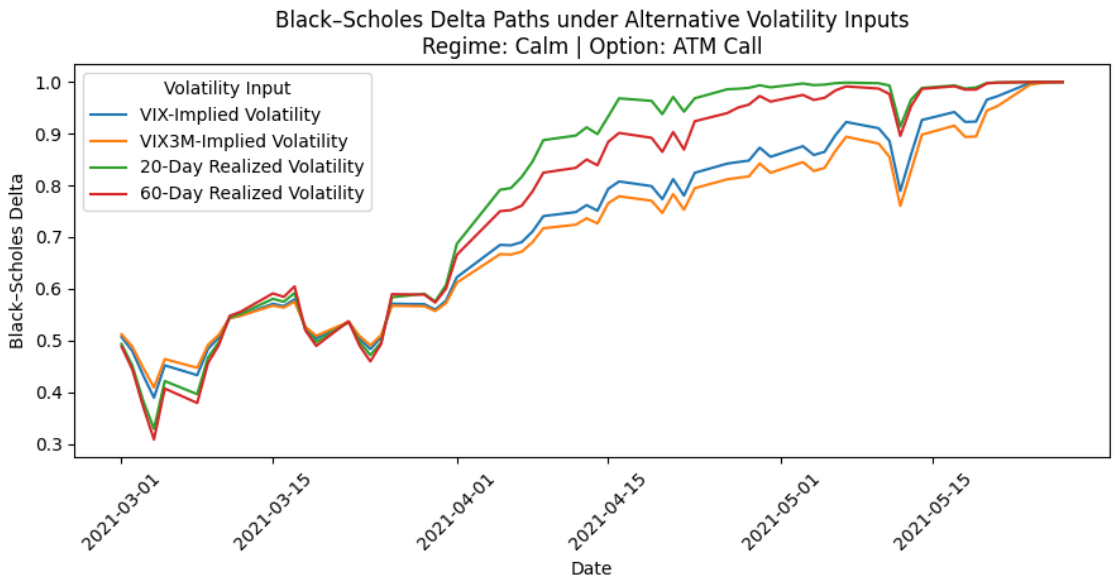
Volatility Input	Variable	Construction	Time Variation
VIX-Implied Volatility	iv_inception_vix	30-day implied volatility fixed at inception	Fixed within episode
VIX3M-Implied Volatility	iv_inception_vix3m	3-month implied volatility fixed at inception	Fixed within episode
20-Day Realized Volatility	rv_20d	Rolling realized volatility from daily returns	Updated daily
60-Day Realized Volatility	rv_60d	Rolling realized volatility from daily returns	Updated daily

[13]:
```
formal_title = (
    "Black-Scholes Delta Paths under Alternative Volatility Inputs\n"
    f"Regime: {sample_info['regime'].title()} | "
    f"Option: {sample_info['moneyness']} {sample_info['option_type'].title()}"
)

plt.figure(figsize=(9, 4.8))
for method in sample["vol_method"].unique():
    sub = sample[sample["vol_method"] == method]
    plt.plot(sub["date"], sub["delta"], label=vol_label_map.get(method, method))

plt.title(formal_title)
plt.xlabel("Date")
plt.ylabel("Black-Scholes Delta")
plt.legend(title="Volatility Input")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig(FIG_DIR / "fig_delta_paths.png", dpi=300, bbox_inches="tight")
```

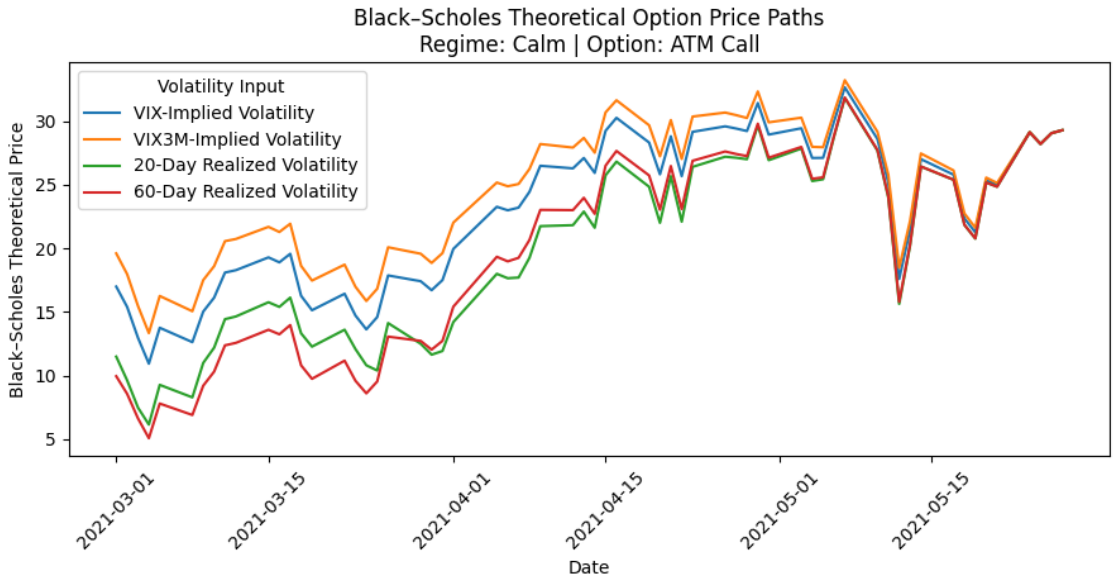
```
plt.show()
plt.close()
```



[14]:
```
formal_title = (
    "Black-Scholes Theoretical Option Price Paths\n"
    f"Regime: {sample_info['regime'].title()} | "
    f"Option: {sample_info['moneyness']} {sample_info['option_type'].title()}"
)

plt.figure(figsize=(9, 4.8))
for method in sample["vol_method"].unique():
    sub = sample[sample["vol_method"] == method]
    plt.plot(sub["date"], sub["bs_price"], label=vol_label_map.get(method,␣
 ↪method))

plt.title(formal_title)
plt.xlabel("Date")
plt.ylabel("Black-Scholes Theoretical Price")
plt.legend(title="Volatility Input")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig(FIG_DIR / "fig_price_paths.png", dpi=300, bbox_inches="tight")
plt.show()
plt.close()
```



[20]:
```
diagnostics_table = (
    bs_panel.groupby("vol_method")
    .agg(
        Obs=("delta", "count"),
        MeanSigma=("sigma", "mean"),
        MeanDelta=("delta", "mean"),
        MinDelta=("delta", "min"),
        MaxDelta=("delta", "max")
    )
    .reset_index()
)

label_map_short = {
    "iv_inception_vix": "VIX IV",
    "iv_inception_vix3m": "VIX3M IV",
    "rv_20d": "RV 20d",
    "rv_60d": "RV 60d"
}

diagnostics_table["vol_method"] = diagnostics_table["vol_method"].
 ↪map(label_map_short)
diagnostics_table = diagnostics_table.rename(columns={"vol_method": "Vol␣
 ↪Input"})

for col in ["MeanSigma", "MeanDelta", "MinDelta", "MaxDelta"]:
    diagnostics_table[col] = diagnostics_table[col].round(4)
```

```
diagnostics_table
fig, ax = plt.subplots(figsize=(9, 2.4))
ax.axis("off")

tbl = ax.table(
    cellText=diagnostics_table.values,
    colLabels=diagnostics_table.columns,
    loc="center",
    cellLoc="center"
)

tbl.auto_set_font_size(False)
tbl.set_fontsize(11)
tbl.scale(1.05, 1.65)

plt.tight_layout()
plt.savefig(FIG_DIR / "table_model_diagnostics.png", dpi=300,␣
 ↪bbox_inches="tight")
plt.show()
```

Vol Input	Obs	MeanSigma	MeanDelta	MinDelta	MaxDelta
VIX IV	741	0.1935	0.0275	-1.0	1.0
VIX3M IV	741	0.2199	0.0325	-1.0	1.0
RV 20d	741	0.2609	0.0743	-1.0	1.0
RV 60d	624	0.2184	0.0686	-1.0	1.0

[16]:
```
report_summary = (
    bs_panel.groupby(["regime", "moneyness", "option_type", "vol_method"])
    .agg(
        mean_sigma=("sigma", "mean"),
        mean_delta=("delta", "mean"),
        sd_delta=("delta", "std"),
        mean_bs_price=("bs_price", "mean"),
        n_obs=("delta", "count")
    )
    .reset_index()
)

report_summary["vol_method_label"] = report_summary["vol_method"].
 ↪map(vol_label_map)
report_summary.to_csv(TABLE_DIR / "bs_delta_summary.csv", index=False)
```

```python
display(report_summary.head(20))
```

```
     regime moneyness option_type          vol_method  mean_sigma  mean_delta ␣
↪sd_delta  mean_bs_price  n_obs  \
0      calm       ATM        call    iv_inception_vix    0.233500    0.723634  0.
↪180592      22.775460     63
1      calm       ATM        call  iv_inception_vix3m    0.267800    0.709678  0.
↪168839      24.255942     63
2      calm       ATM        call              rv_20d    0.139658    0.787731  0.
↪226650      19.800164     63
3      calm       ATM        call              rv_60d    0.148683    0.766657  0.
↪218295      19.565579     63
4      calm       ATM         put    iv_inception_vix    0.233500   -0.274767  0.
↪179677       7.894279     63
5      calm       ATM         put  iv_inception_vix3m    0.267800   -0.288724  0.
↪167923       9.374761     63
6      calm       ATM         put              rv_20d    0.139658   -0.210670  0.
↪225783       4.918983     63
7      calm       ATM         put              rv_60d    0.148683   -0.231745  0.
↪217405       4.684398     63
8      calm       OTM        call    iv_inception_vix    0.233500    0.514911  0.
↪166723      11.378959     63
9      calm       OTM        call  iv_inception_vix3m    0.267800    0.518135  0.
↪148632      13.075376     63
10     calm       OTM        call              rv_20d    0.139658    0.534474  0.
↪238458       7.138312     63
11     calm       OTM        call              rv_60d    0.148683    0.508809  0.
↪235664       7.477061     63
12  highvol       ATM        call    iv_inception_vix    0.196300    0.142903  0.
↪172513       3.578917     62
13  highvol       ATM        call  iv_inception_vix3m    0.242100    0.170132  0.
↪174308       5.053613     62
14  highvol       ATM        call              rv_20d    0.270310    0.185365  0.
↪153122       4.586958     62
15  highvol       ATM        call              rv_60d    0.251279    0.172608  0.
↪166633       4.625823     62
16  highvol       ATM         put    iv_inception_vix    0.196300   -0.855329  0.
↪173407      47.174011     62
17  highvol       ATM         put  iv_inception_vix3m    0.242100   -0.828099  0.
↪175238      48.648707     62
18  highvol       ATM         put              rv_20d    0.270310   -0.812866  0.
↪154071      48.182052     62
19  highvol       ATM         put              rv_60d    0.251279   -0.825623  0.
↪167565      48.220917     62
```

```
                vol_method_label
0        VIX-Implied Volatility
1      VIX3M-Implied Volatility
2    20-Day Realized Volatility
3    60-Day Realized Volatility
4        VIX-Implied Volatility
5      VIX3M-Implied Volatility
6    20-Day Realized Volatility
7    60-Day Realized Volatility
8        VIX-Implied Volatility
9      VIX3M-Implied Volatility
10   20-Day Realized Volatility
11   60-Day Realized Volatility
12       VIX-Implied Volatility
13     VIX3M-Implied Volatility
14   20-Day Realized Volatility
15   60-Day Realized Volatility
16       VIX-Implied Volatility
17     VIX3M-Implied Volatility
18   20-Day Realized Volatility
19   60-Day Realized Volatility
```

1.8 8. Save deliverables

[21]:
```
output_parquet = DATA_PATH / "bs_delta_panel.parquet"
output_csv = DATA_PATH / "bs_delta_panel.csv"
episode_summary_csv = TABLE_DIR / "episode_summary.csv"
summary_csv = TABLE_DIR / "bs_delta_summary.csv"
readme_path = DATA_PATH / "bs_delta_panel_README.md"
manifest_path = DATA_PATH / "modeling_outputs_manifest.txt"

bs_panel.to_parquet(output_parquet, index=False)
bs_panel.to_csv(output_csv, index=False)
episode_summary.to_csv(episode_summary_csv)
report_summary.to_csv(summary_csv, index=False)

readme_text = f"""# Black-Scholes Delta Panel

Generated by `black_scholes_model.ipynb`.

## Main handoff files
- `bs_delta_panel.parquet`
- `bs_delta_panel.csv`

## Unit of observation
One row = one option episode × one trading day × one volatility method.
```

```
## Key columns
- `episode_id`: option experiment identifier.
- `regime`: market regime label.
- `date`: trading date.
- `S`: SPY traded close price.
- `K`: option strike.
- `tau`: remaining time to maturity in years.
- `r`: daily risk-free rate input, decimal.
- `q`: continuous dividend-yield proxy.
- `sigma`: volatility input used for Black-Scholes.
- `bs_price`: theoretical Black-Scholes option value.
- `delta`: Black-Scholes delta.
- `option_type`: `call` or `put`.
- `moneyness`: ATM/OTM label from the episode file.
- `vol_method`: volatility specification. Current values:␣
 ↪{sorted(bs_panel['vol_method'].unique().tolist())}.

## Report resources saved
Figures in `report_figures/`
Tables in `report_tables/`

## Handoff note
Use `delta` as the hedge ratio for the rebalancing simulation.
Filter by `vol_method` to compare hedging performance under different␣
 ↪volatility assumptions.
"""

manifest_text = "\n".join([
    "MODEL HANDOFF FILES",
    str(output_parquet),
    str(output_csv),
    str(readme_path),
    "",
    "REPORT FIGURES",
    str(FIG_DIR / "fig_delta_paths.png"),
    str(FIG_DIR / "fig_price_paths.png"),
    str(FIG_DIR / "table_volatility_inputs.png"),
    str(FIG_DIR / "table_model_diagnostics.png"),
    "",
    "REPORT TABLE CSVS",
    str(TABLE_DIR / "table_volatility_inputs.csv"),
    str(TABLE_DIR / "table_model_diagnostics.csv"),
    str(TABLE_DIR / "bs_delta_summary.csv"),
    str(TABLE_DIR / "episode_summary.csv"),
])

readme_path.write_text(readme_text, encoding="utf-8")
```

```
manifest_path.write_text(manifest_text, encoding="utf-8")

print("Saved files:")
for p in [
    output_parquet, output_csv, episode_summary_csv, summary_csv,
    readme_path, manifest_path,
    FIG_DIR / "fig_delta_paths.png",
    FIG_DIR / "fig_price_paths.png",
    FIG_DIR / "table_volatility_inputs.png",
    FIG_DIR / "table_model_diagnostics.png",
]:
    print(" -", p)
```

```
Saved files:
 - C:\Users\elain\Downloads\5010-Project\data\output\bs_delta_panel.parquet
 - C:\Users\elain\Downloads\5010-Project\data\output\bs_delta_panel.csv
 - C:\Users\elain\Downloads\5010-
Project\data\output\report_tables\episode_summary.csv
 - C:\Users\elain\Downloads\5010-
Project\data\output\report_tables\bs_delta_summary.csv
 - C:\Users\elain\Downloads\5010-Project\data\output\bs_delta_panel_README.md
 -
C:\Users\elain\Downloads\5010-Project\data\output\modeling_outputs_manifest.txt
 - C:\Users\elain\Downloads\5010-
Project\data\output\report_figures\fig_delta_paths.png
 - C:\Users\elain\Downloads\5010-
Project\data\output\report_figures\fig_price_paths.png
 - C:\Users\elain\Downloads\5010-
Project\data\output\report_figures\table_volatility_inputs.png
 - C:\Users\elain\Downloads\5010-
Project\data\output\report_figures\table_model_diagnostics.png
```

[18]:
```
check = pd.read_parquet(DATA_PATH / "bs_delta_panel.parquet")
print("Reloaded shape:", check.shape)
display(check.head())

print("\nSaved figure files:")
for p in sorted(FIG_DIR.glob("*.png")):
    print(" -", p.name)

print("\nSaved table files:")
for p in sorted(TABLE_DIR.glob("*.csv")):
    print(" -", p.name)
```

```
Reloaded shape: (2847, 19)

       episode_id regime       date           S      K       tau        r       ␣
↪ q   sigma   bs_price     delta  \
```

```
0  SPY_calm_C_ATM   calm 2021-03-01  389.579987  390.0  0.241096  0.00028  0.
 ↪013122  0.2335  16.989992  0.506741
1  SPY_calm_C_ATM   calm 2021-03-02  386.540009  390.0  0.238356  0.00035  0.
 ↪013122  0.2335  15.399808  0.479406
2  SPY_calm_C_ATM   calm 2021-03-03  381.420013  390.0  0.235616  0.00035  0.
 ↪013122  0.2335  12.971057  0.432680
3  SPY_calm_C_ATM   calm 2021-03-04  376.700012  390.0  0.232877  0.00028  0.
 ↪013122  0.2335  10.937231  0.389266
4  SPY_calm_C_ATM   calm 2021-03-05  383.630005  390.0  0.230137  0.00028  0.
 ↪013122  0.2335  13.758139  0.451908

  option_type cp_flag moneyness        vol_method                           ␣
 ↪q_source inception_date expiration  \
0        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
1        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
2        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
3        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28
4        call       C       ATM  iv_inception_vix  div_amount_in_window / S0 /␣
 ↪T_years     2021-03-01 2021-05-28

           S0
0  389.579987
1  389.579987
2  389.579987
3  389.579987
4  389.579987


Saved figure files:
 - fig_delta_paths.png
 - fig_price_paths.png
 - table_model_diagnostics.png
 - table_volatility_inputs.png

Saved table files:
 - bs_delta_summary.csv
 - episode_summary.csv
 - table_model_diagnostics.csv
 - table_volatility_inputs.csv
```

1.9 9. Suggested team handoff message

I finished the Black–Scholes modeling layer. The main handoff file is `data/output/bs_delta_panel.parquet`, with one row per episode × trading day × volatility

method. It includes theoretical option price, remaining maturity, risk-free rate, volatility input, and delta. Use `delta` directly for the rebalancing simulation, and compare results across `vol_method`. Formal figures and tables for the report are saved under `data/output/report_figures/` and `data/output/report_tables/`.